Practice Problems Unit 5
AP Calculus

Name ANSWERS

1.) –12

2.) $\frac{5}{6}$

3.) (a) 6
(b) $y + 2 = 6(x - 4)$

4.) (a) $x = -2$, $x = 0$, $x = 1$ $f''(x) = 0$ and changes signs.
(b) Concave up: $(-2, 0)$, $(1, 3)$, $(3, +\infty)$
Concave down: $(-\infty, -2)$, $(0, 1)$

5.) (a) $x = 0$, $x = -5$, $x = 7$
(b) $x = -5$ is a relative maximum $x = 7$ is a relative minimum

6.) increasing: (–∞, 1] [3, ∞) because $f'(x) \geq 0$ on these intervals
decreasing: [1, 3] because $f'(x) \leq 0$ on this interval

concave up: (–3, –1) and (2, ∞) because $f'(x)$ is increasing on these intervals
concave down: (–∞, –3) and (–1, 2) because $f'(x)$ is decreasing on these intervals

x-values of points of inflection:
$x = -3$ and $x = 2$ because $f'(x)$ has a horizontal tangent line at these points and $f'(x)$changes from decreasing to increasing at these values.
$x = -1$ because $f'(x)$ has a horizontal tangent line at these points at this point and $f'(x)$ changes from increasing to decreasing at this value.

7.) moving right: (0, 3) because $v(t) > 0$ on this interval
moving left: (3, 6) because $v(t) < 0$ on this interval
Acceleration is positive: (0, 2) and (5, 6) because $v(t)$ is increasing
Acceleration is negative: (2, 5) because $v(t)$ is decreasing
speeding up: (0, 2) because $v(t) > 0$ and $v(t)$ is increasing
(3, 5) because $v(t) < 0$ and $v(t)$ is decreasing
slowing down: (2, 3) because $v(t) > 0$ and $v(t)$ is decreasing
(5, 6) because $v(t) < 0$ and $v(t)$ is increasing

8.)

x	A	B	C	D	E	F	G	H	I	J
$f(x)$	$-$	$+$	$+$	0	$+$	$+$	$+$	$-$	$-$	0
$f'(x)$	$+$	$+$	$-$	0	$+$	0	$-$	0	$+$	$+$
$f''(x)$	$-$	$-$	0	$+$	0	$-$	$-$	$+$	$+$	$+$